

RF Field Strength Meter



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Goal

Wideband detection in
VHF/UHF bands using
cost-effective BJTs.

0.003 MHz		Very Low Frequency (VLF)
0.03 MHz		Low Frequency (LF)
0.3 MHz		Medium Frequency (MF)
3 MHz		High Frequency (HF)
30 MHz		Very High Frequency (VHF)
300 MHz		Ultra High Frequency (UHF)
3 000 MHz		Super High Frequency (SHF)
30 000 MHz		Extra High Frequency (EHF)
300 000 MHz		

Requirements

IN60

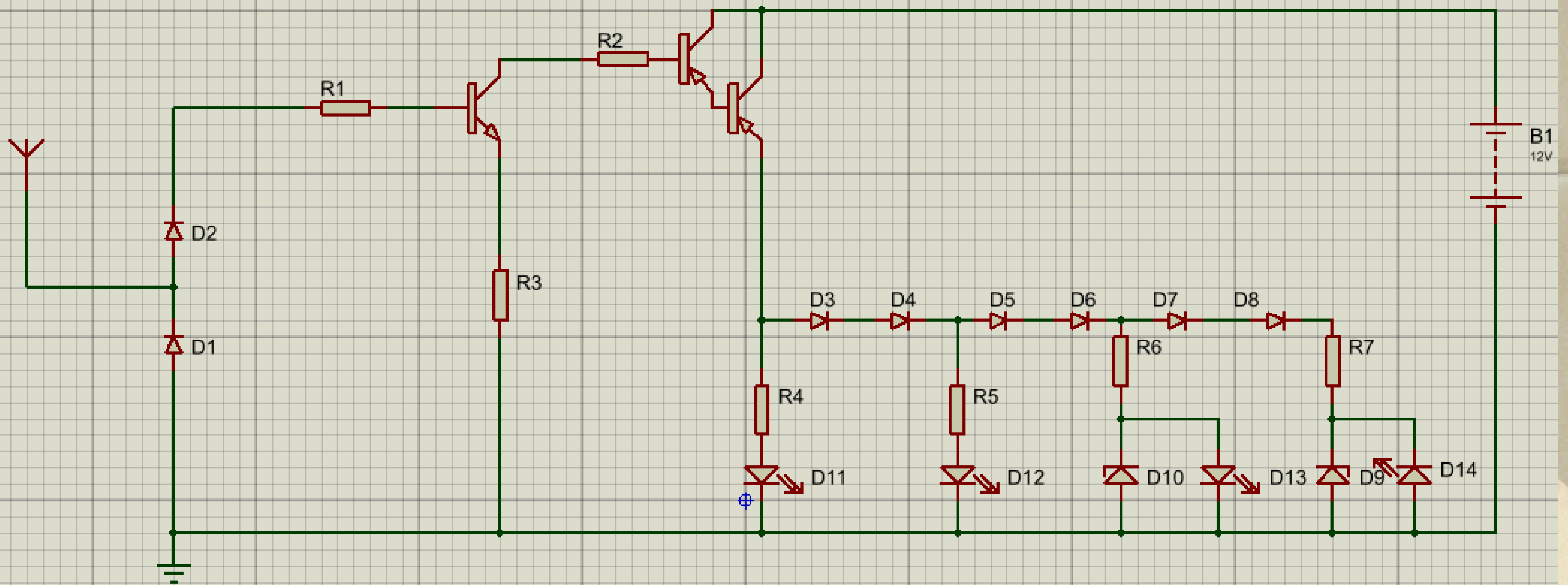
2N4401

1N4148

3.3 Zener

BC516

Schematic



What about the
antenna?

Wavelength

C: Light Speed Constants = 300×10^6 m/s

f: My Targeted Frequency = 443 MHz

$$\lambda = \frac{c}{f}$$

Antenna Length

C: Light Speed Constants = 300×10^6 m/s

f: My Targeted Frequency = 443 MHz

$$L = \frac{\lambda}{4}$$

Test

Thank you for listening

